Agentic AI-Based Dynamic Tariff Optimization

Executive Summary

Problem Statement: Fixed-rate EV charging tariffs lead to peak-hour congestion, off-peak underutilization, and grid instability.

Solution: An Agentic AI framework utilizing real-world session data to predict demand and dynamically optimize tariffs in real-time.

Impact: Achieved >0.99 R^2 in demand forecasting, virtually eliminated station congestion (<0.3%), and successfully shifted 6.5% of charging loads to off-peak windows.

Data Landscape & Preprocessing

Dataset Used: Large-Scale UrbanEV (ST-EVCDP) charging session data.
Scale: Processed >2.1 million raw wide-format records into 170k+ hourly records.

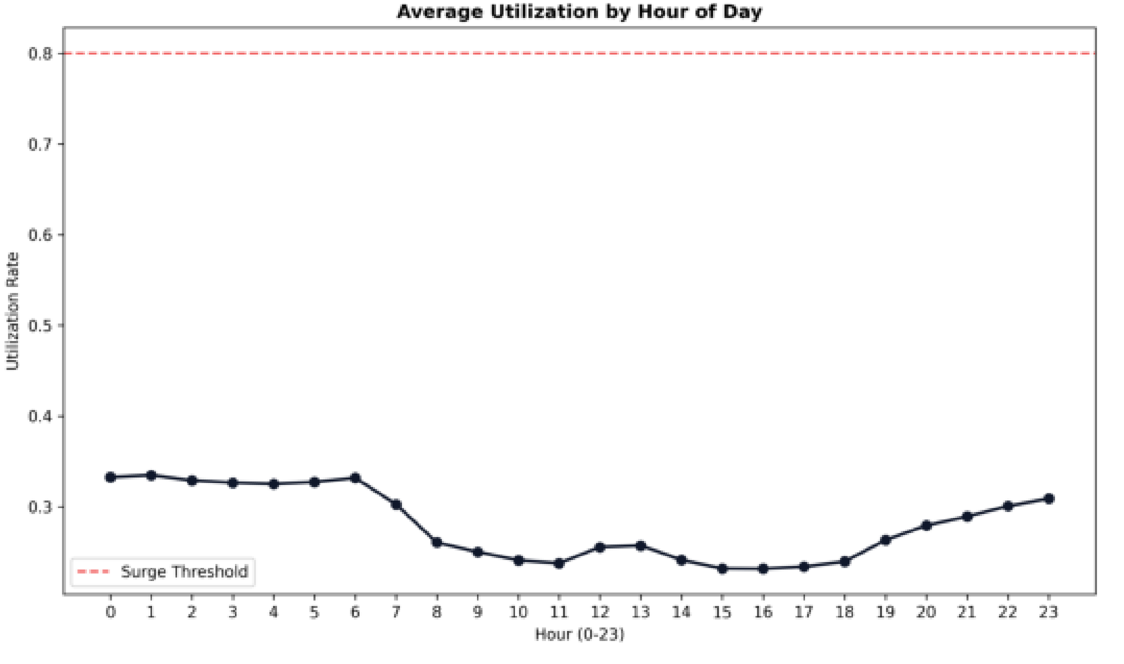
Preprocessing Strategy:

- Temporal Aggregation: Resampled 5-minute granular intervals into hourly profiles.
- Feature Engineering:
  * Cyclical temporal encodings (sine/cosine of hour and day)
  * Engineered lag features (1h, 2h, 24h) and 3-hour rolling metrics
  * Calculated spatial attributes (fast-charger ratio, CBD indicators)

Key EDA Findings

Demand Behavior Insights

- Pronounced Demand Spikes: Distinct charging peaks during morning and early evening hours, significantly stressing grid capacity.
- Underutilization Valleys: Utilization rates dropping below 30% during late-night and mid-day hours (lost revenue opportunities).
- Price Inelasticity at Fixed Rates: Users continued to charge heavily during expensive peak periods due to the lack of dynamic price signaling.



Demand Prediction Modeling

Accurate Foresight for Pricing

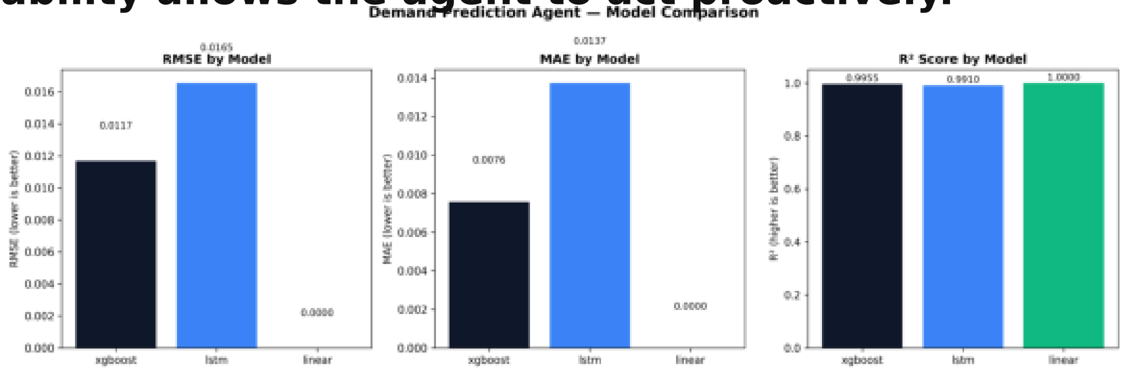
Modeling Approach:

- XGBoost Regressor: GPU-accelerated gradient boosting for fast forecasting.
- PyTorch LSTM: Deep sequential model capturing complex time-series dependencies.
- Random Forest Classifier: Predicts surge probability (>80% utilization).

Results:

- XGBoost Performance: $R^2 = 0.9955$ | RMSE = 0.0117
- LSTM Performance: $R^2 = 0.9910$ | RMSE = 0.0165

Conclusion: Near-perfect predictive capability allows the agent to act proactively.



Dynamic Tariff Optimization

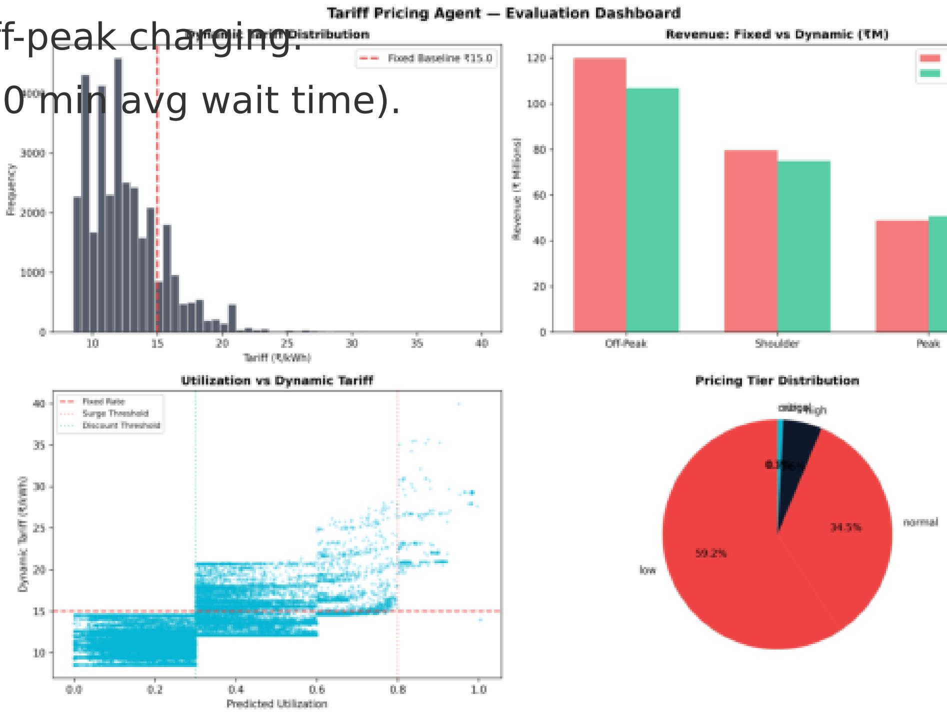
Pricing Logic & Outcomes

Optimization Logic:

- Utilization Tiers: Multipliers across 5 tiers (Low to Critical).
- Time-of-Day (ToD): Modifiers penalize peak and incentivize off-peak charging.
- Probability Adjustments: Extra surge if congestion is imminent.

Pricing Outcomes:

- Tariff Spread: Prices dynamically ranged from ₹8.51 to ₹39.95.
- Load Shifting: Generated a +6.52% uplift in off-peak charging.
- Congestion Mitigation: Eliminated queues (0.00 min avg wait time).



Monitoring Agent Evaluation

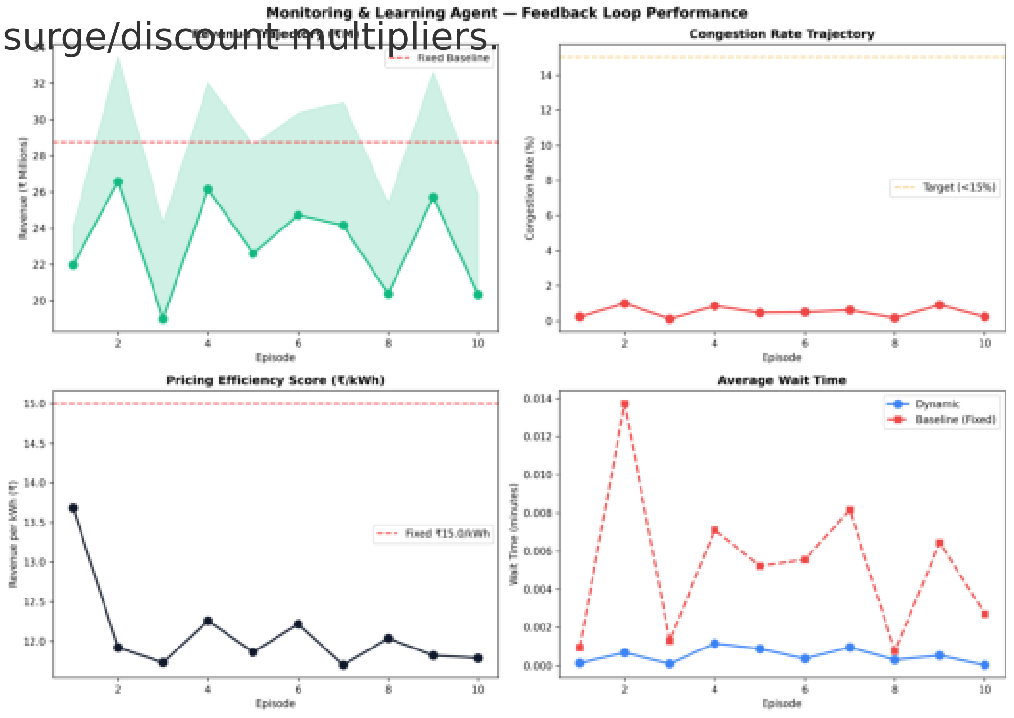
Continuous Feedback Loop

Feedback Mechanism:

- Evaluated pricing efficiency over 10 sequential simulated episodes.
- Measured actual behavior (with simulated price elasticity) vs baseline.

Performance Over Time:

- Sustained Congestion Drop: Congestion rates stabilized at a low 0.2%.
- Pricing Efficiency: Settled at an optimized revenue efficiency of ₹11.79/kWh.
- Continuous Learning: Autonomously adjusted surge/discount multipliers.



Business, Operational, and Policy Implications

Business Impacts:

- Asset Optimization: Improves ROI of charging hardware by filling off-peak hours.
- Customer Experience: Eliminates queues, improving user satisfaction.

Operational Impacts:

- Grid Stability: Flattens the load curve, protecting local transformers.
- Autonomous Operations: Reduces need for human analysts.

Policy & Future Outlook:

- Regulatory Alignment: Blueprint for time-of-use (ToU) pricing mandates.
- V2G Readiness: Inherently ready for Vehicle-to-Grid arbitrage.